

Corporate Finance

Study notes from zero: Time Value of Money, Present Value, NPV, Annuities, Perpetuities & Bonds

Built from your lecture slides • Plain-English explanations • Diagrams • Worked examples • 20 practice problems with solutions

How to use these notes

Your slides are theory-heavy because finance is really **one idea repeated many times**. Once you see that idea, everything else (NPV, bonds, annuities…) is just that idea applied to different situations.

The ONE big idea of this whole subject

A rupee today is worth more than a rupee tomorrow. (Because today's rupee can be put in a bank and earn interest.) So to compare money at different dates, we must move it to the *same date*. Moving forward = **compounding**. Moving backward = **discounting**.

Roadmap (follows your slide order)

Part	Topic	What you'll be able to do
1	Setting the stage: markets, loans, bonds	Understand the vocabulary and the 'perfect market' assumptions
2	Returns	Compute rate of return, capital gain, dividend yield, basis points
3	Future value & compounding	Grow money forward in time
4	Present value & discounting	Shrink future money back to today
5	Net Present Value (NPV)	Decide whether a project is worth doing
6	Choice sets (saving & borrowing)	Draw the budget line across two periods; inflation & real rate
7	Asset markets & no-arbitrage	Why prices must equal present values; forest example; continuous compounding
8	Perpetuities & annuities	Shortcut formulas for long streams of payments
9	Applications: lottery & bonds	Price a bond; premium vs discount
10	Formula cheat-sheet	One page to revise from
11	Practice problems + solutions	Test yourself

Tip: try to cover the 'Example' boxes and solve them yourself before reading the answer. Then try the practice problems in Part 11 **without peeking** at solutions.

Note about your two files

FF_2.pdf and **CF_2A_v2.pdf** are the same lecture (Chapter 2: Present Value). They differ on **one slide**: the rearranged inflation budget line. The newer file (CF_2A_v2) has the correct version: $c_2 = -[(1+r)/(1+\pi)]\cdot c_1 + [m_1(1+r)+m_2]/(1+\pi)$. These notes use the corrected one. (The older file shows $(1+\pi)(m_1/(1+r)+m_2)$, which does not follow from the line above it.)

PART 1

Setting the stage

Why we need present value, and the rules of the game

1.1 Why does finance exist?

Some people have money today and want more later (**lenders / investors**). Others need money today to do something useful (**borrowers**, e.g. firms). Finance is the study of how money moves between them across time.

Deciding *which projects a firm should spend money on* is called **capital budgeting**. The main tool for this is **Net Present Value (NPV)**, which is the destination of these notes.

> **The NPV rule in one line**
>
> Convert all future cash flows (money in **and** out) into today's money, add them up (including today's cost). **If the total is positive → do the project. If negative → reject it.**

1.2 Perfect market (the 'simple world' we start in)

Code	Assumption	In plain English
P1	No taxes	Nobody loses money to the government
P2	No transaction costs	Buying/selling is free (no brokerage fees)
P3	Same information & opinions	Everyone sees the same facts and agrees on them
P4	Many buyers and sellers	No single person can move the price

The slides add **three extra simplifications** (not part of 'perfect market', just to make maths easy):

Code	Simplification	Meaning
E1	Same interest rate each period	One rate r for all years
E2	No inflation	Prices of goods don't rise (relaxed later in Part 6)
E3	No risk or uncertainty	All future cash flows are 100% certain

1.3 Loans and bonds

- **Loan**: a promise by a borrower to pay fixed amounts at fixed future dates, usually in exchange for cash today.
- **Maturity**: the final date when the loan is fully paid off.
- **Bond**: a special kind of loan. If *you buy* a bond you are **lending** money. If a *firm issues/sells* a bond it is **borrowing**.
- Loans and bonds are also called **fixed income securities** (the payments are fixed in advance).

> **Example: Bond as a loan**
>
> You give a company ₹1,000 today. It promises you ₹1,100 in one year. You have 'bought a bond'; the company has 'issued a bond'.

PART 2

Returns

How much did my investment earn?

2.1 Cash return and net return

We label time with **t**: **t = 0 is today**, t = 1 is next period (a year, say), and **T** usually means the last period. $\mathbf{C_t}$ is the cash at time t.

- **Net return** = what you get back − what you put in = $C_1 - C_0$.
- **Rate of return** = net return as a % of what you put in.

RATE OF RETURN

$$r_1 = \frac{C_1 - C_0}{C_0}$$

Example: Simple return

You pay ₹100 today and get ₹120 next year. Net return = 120 − 100 = ₹20. Rate of return = 20/100 = **20%**.

2.2 Investments with interim payments (dividends, coupons, rent)

Stocks pay **dividends**, bonds pay **coupons**, houses give **rent**. These payments arrive *before* the end and count as part of your return.

Example: Share with a dividend

You buy a share for ₹92. It pays a ₹5 dividend, and is then worth ₹110.
Rate of return = (110 + 5 − 92) / 92 = 23/92 = **25%**.

2.3 Splitting the return into two parts

RETURN = CAPITAL GAIN + DIVIDEND YIELD

$$r_1 = \frac{C_1 - C_0}{C_0} + \frac{\text{Dividends}}{C_0}$$

Part	Meaning	In the ₹92 share example
Capital gain	Price change only (ignores dividends)	110 − 92 = 18 → 18/92 = **19.6%**
Dividend yield	Interim payments ÷ purchase price	5/92 = **5.4%**
Total return	Add both	19.6% + 5.4% = **25%** ✔

You can have a **negative capital gain and still a positive total return** if the dividends are big enough (e.g. price falls ₹92 → ₹90 but you receive ₹5: return = (90+5−92)/92 = +3.3%).

2.4 Percentage points and basis points

Is 10% 'more than 5%' by 5% or by 100%? Both statements are true in different senses, which is confusing. So finance uses clear units:

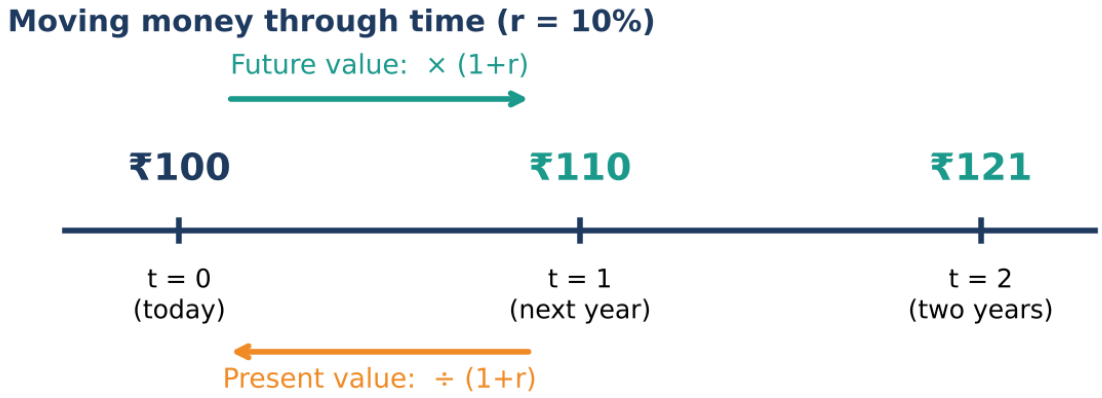
Unit	Size	5% → 10% change is…
1 percentage point	1%	5 points
1 basis point (bp)	0.01%	500 bp

- Rule: **100 basis points = 1 percentage point**. Central banks and bond traders talk in basis points all the time.
- (Also true: 10% is 100% *relatively* higher than 5%, because it doubles. But 'points' avoids confusion.)

PART 3

Future value & compounding

How much will my money grow to?



3.1 Future value (FV)

Rearranging the rate-of-return formula gives the amount you will have after one period:

FUTURE VALUE, ONE PERIOD

$$C_1 = C_0(1 + r)$$

Example: ₹100 at 10%

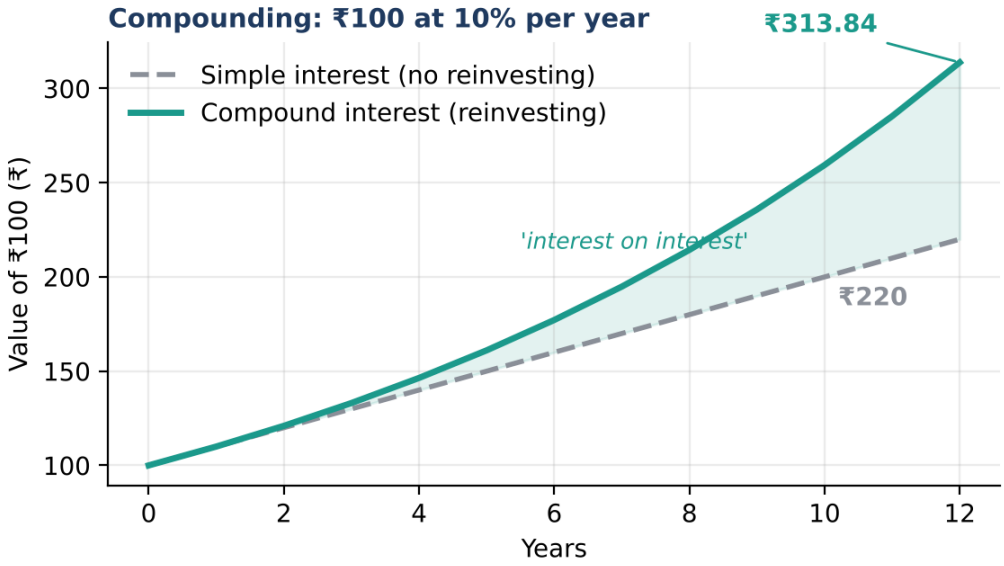
C_1 = 100 × 1.10 = **₹110**. We say ₹110 is the **future value** of ₹100 today.

3.2 Compounding: interest on interest

If you reinvest and keep earning 10% every year, after 2 years you have 110 × 1.10 = ₹121. The 2-year return is **21%, not 20%**. The extra ₹1 is the interest earned on the first year's interest. This is **compound interest**.

t-YEAR COMPOUNDING

$$1 + r_t = (1 + r)^t \qquad \text{so} \qquad FV = C_0\,(1 + r)^t$$



Example: ₹100 for 12 years at 10%

FV = 100 × $(1.1)^{12}$ ≈ **₹313.84**. (Simple interest would give only 100 + 12×10 = ₹220.)

3.3 Three classic 'reverse' questions

Question	Set-up	Answer
Constant yearly rate r that gives a 2-year return of 50%?	$1.50 = (1+r)^2 \rightarrow r = \sqrt{1.5} - 1$	**22.47%** (not 25%! check: $1.25^2 - 1 = 56.25\%$)
Years to double money at 3%?	$(1.03)^x = 2 \rightarrow x = \log 2 / \log 1.03$	**≈ 23.45 years**
Half-year return if yearly rate is 5%?	$(1+r_{0.5})^2 = 1.05 \rightarrow r = \sqrt{1.05} - 1$	**≈ 2.47%** (not 2.5%)

Shortcut: Rule of 72

Years to double ≈ 72 ÷ (rate in %). At 3%: 72/3 = 24 years (exact answer 23.45). At 8%: about 9 years. This is a quick mental check (not in your slides, but handy).

PART 4

Present value & discounting

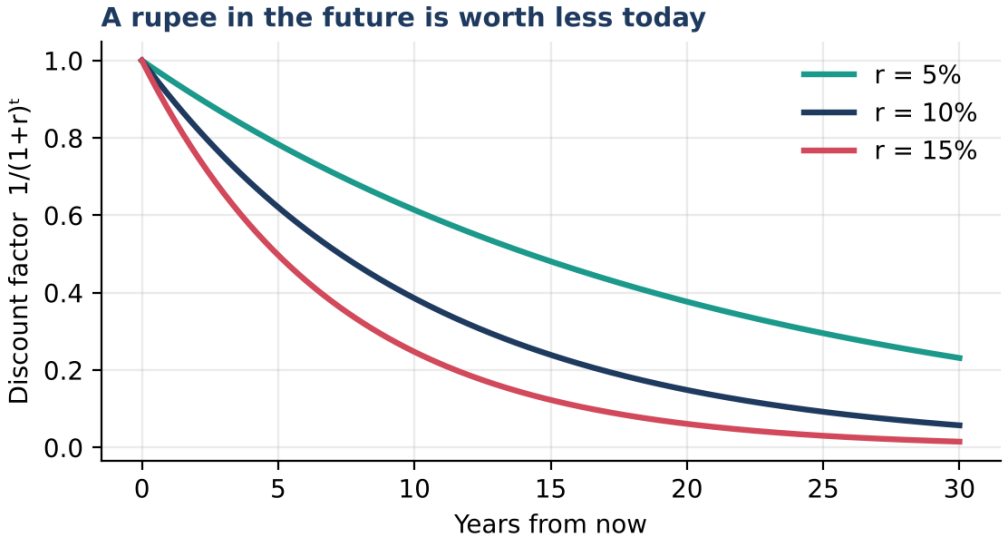
What is future money worth today?

This is the **mirror image** of future value. Instead of asking 'what will ₹100 become?', we ask 'how much must I put in today to get ₹100 later?'

PRESENT VALUE

$$PV = C_0 = \frac{C_1}{1+r} \qquad PV = \frac{C_t}{(1+r)^t}$$

- **Discounting** = reverse of compounding.
- **Discount rate** r = the rate of return available elsewhere. Also called the **cost of capital** (or opportunity cost).
- **Discount factor** = $1/(1+r)^t$. Multiply a future cash flow by it to get present value.



Read the chart: at 10%, ₹1 received in 10 years is worth only about 39 paise today. The higher the rate and the longer the wait, the smaller the present value.

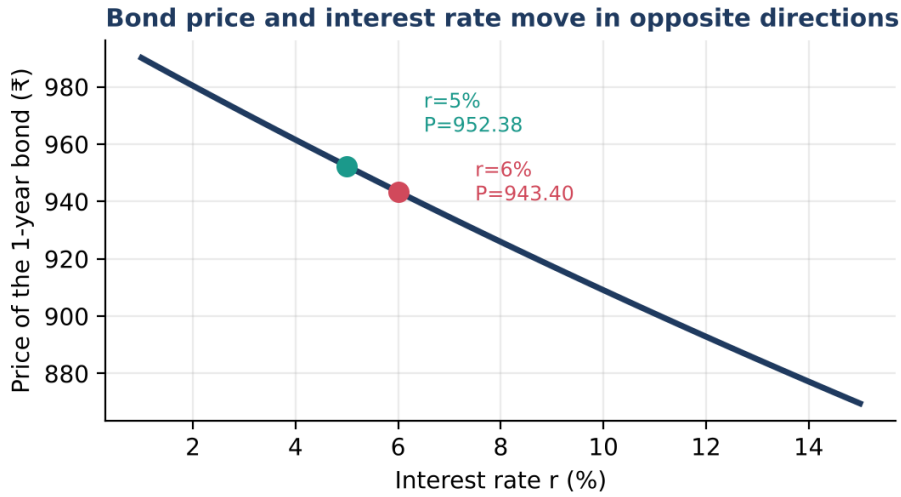
Example: PV of ₹100 next year at 10%

PV = 100 / 1.10 = **₹90.91**. Discount factor = 1/1.10 ≈ 0.91.
Two years: PV = 100 / $(1.1)^2$ = **₹82.64** (discount factor ≈ 0.83).

Example: Is the project a good deal?

A project costs ₹90.91 and returns ₹100 next year.
• If the market rate is **12%**: PV = 100/1.12 = 89.29 < 90.91 → **not** a good deal (you pay more than it's worth).
• If the market rate is **9%**: PV = 100/1.09 = 91.74 > 90.91 → **good** deal.

4.1 Bond prices and interest rates move in opposite directions



A bond paying ₹1,000 in one year is worth 1000/1.05 = **₹952.38** at 5%, but only 1000/1.06 = **₹943.40** at 6%. **When interest rates rise, fixed-payment bond prices fall** (and vice-versa).

PART 5

Net Present Value (NPV)

The most important tool in corporate finance

Why present value is so useful: you **cannot** add ₹10 next year to ₹8 in five years (they are different 'currencies' of time). But once both are converted to *today's rupees*, you can add, subtract and compare freely.

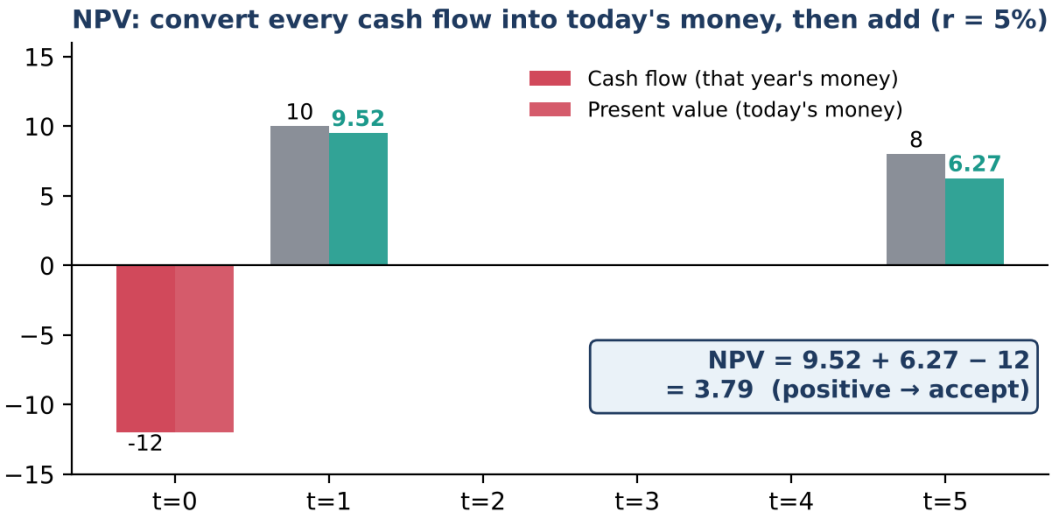
NPV FORMULA (C_0 is usually negative: the upfront cost)

$$NPV = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \cdots + \frac{C_T}{(1+r)^T}$$

3 steps: ① convert each future cash flow to today's money, ② add them up, ③ subtract the initial cost. 'Net' reminds you to include *all* inflows and outflows.

Decision rule

NPV > 0 → accept | **NPV < 0 → reject** | NPV = 0 → indifferent (you just earn the market rate).



Example: Your slide's project (cost 12; get 10 in year 1 and 8 in year 5; r = 5%)

PV(10) = 10/1.05 = 9.52 PV(8) = 8/1.05^5 = 6.27 NPV = 9.52 + 6.27 − 12 = **+3.79** → accept.

What does NPV = 3.79 actually mean? (two ways to see it)

- **Value added:** The project's future cash is worth 15.79 today but costs only 12. You gain 3.79 in today's money.
- **Versus the bank:** Put 12 in the bank at 5% → 12.60 next year. Take out 10 to match the project's year-1 payment; 2.60 stays invested and grows to 2.60 × 1.05^4 ≈ 3.16 by year 5. The project instead gives 8 in year 5. Since 8 > 3.16, the project beats the bank.

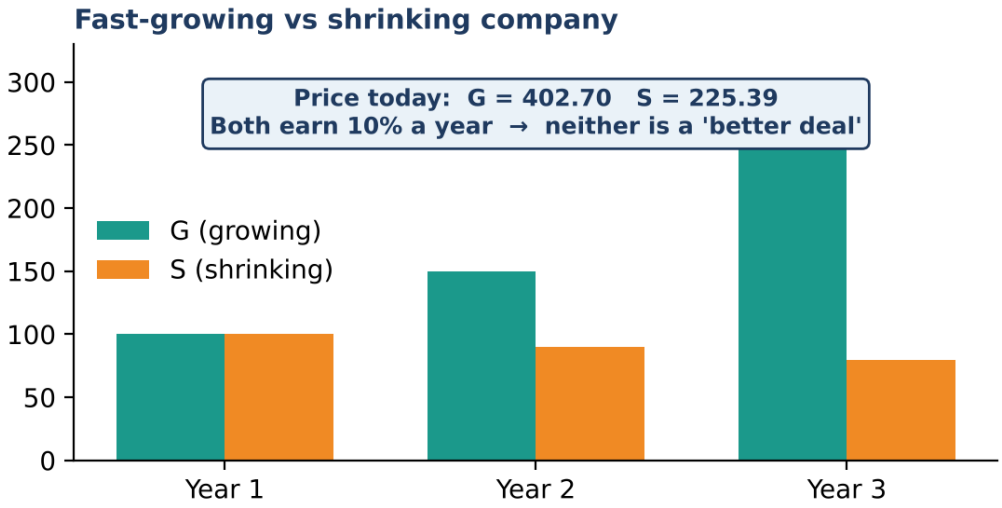
5.1 A longer example (with a clean-up cost)

Example: Project with 5 cash flows

Cost 900 today; +200 in years 1 & 2; +400 in years 3 & 4; clean-up cost −100 in year 5; r = 5%.
NPV = −900 + 200/1.05 + 200/1.05^2 + 400/1.05^3 + 400/1.05^4 − 100/1.05^5 ≈ **+68.15** → accept.

5.2 Does it pay to buy the fast-growing company?

Company **G** (growing) pays 100, 150, 250 over 3 years. Company **S** (shrinking) pays 100, 90, 80. Both face a 10% cost of capital and there is no uncertainty.



	Price today (PV)	Value next year (cash 100 + remaining)	One-year return
G	$100/1.1 + 150/1.1^2 + 250/1.1^3$ = **402.70**	$100 + 150/1.1 + 250/1.1^2$ = 442.98	(442.98 − 402.70)/402.70 = **10%**
S	$100/1.1 + 90/1.1^2 + 80/1.1^3$ = **225.39**	$100 + 90/1.1 + 80/1.1^2$ = 247.93	(247.93 − 225.39)/225.39 = **10%**

Lesson

Market prices **already include** expected growth. G costs more today precisely because it will pay more later. Both earn the same fair 10%, so **neither is a 'better deal'**. Growth alone does not make an investment good; only NPV > 0 does.

PART 6

Choice sets: saving & borrowing over time

Using present value to understand consumer choice

(This part of your slides comes from microeconomics. It shows why PV is natural: people can move spending between periods by saving or borrowing.)

6.1 Warm-up: the ordinary budget set

A consumer has income **m** and buys quantities (x_1, x_2) at prices (p_1, p_2). A bundle is **affordable** if $p_1x_1 + p_2x_2 \leq m$. Bundles that use up *exactly* m form the **budget line**; all affordable bundles form the **budget set**.

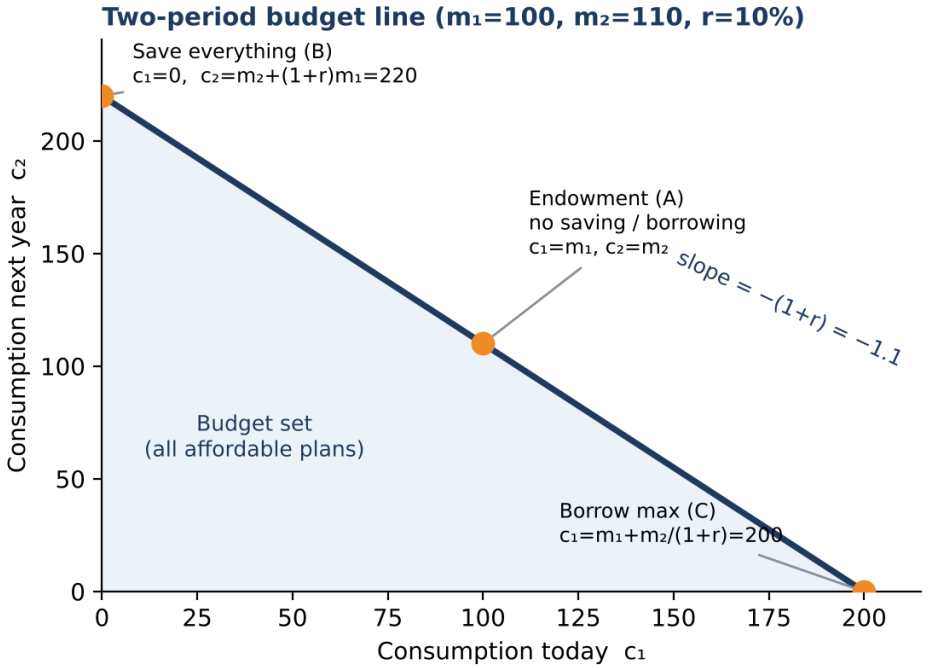
Example: Chai and samosas

Income ₹100, chai ₹10, samosa ₹20. Affordable: 10 chai & 0 samosa, or 0 chai & 5 samosas, or 5 chai & 2.5 samosas... these lie on the line $10x_1 + 20x_2 = 100$.

6.2 Same idea across time

Now the 'goods' are **consumption this year (c_1)** and **consumption next year (c_2)**. You earn m_1 now and m_2 next year; interest rate is r; prices are 1 (for now).

- **No saving/borrowing:** $c_1 = m_1$, $c_2 = m_2$ (point A).
- **Save everything today (c_1=0):** $c_2 = m_2 + (1+r)m_1$ (point B).
- **Borrow the maximum today (c_2=0):** you can repay at most m_2, so borrow b with $b(1+r) = m_2$, so $b = m_2/(1+r)$; $c_1 = m_1 + m_2/(1+r)$ (point C).



INTERTEMPORAL BUDGET CONSTRAINT

$$c_1 + \frac{c_2}{1+r} = m_1 + \frac{m_2}{1+r}$$

In words

Present value of what you consume = present value of what you earn. Every point on the line satisfies this. Slope = −(1+r): giving up ₹1 today gets you (1+r) next year.

Example: Numbers

m_1=100, m_2=110, r=10%. PV of income = 100 + 110/1.1 = **200**. Max c_1 = 200 (borrow ₹100 against next year's pay). Max c_2 = 110 + 1.1×100 = **220**. If you spend c_1=150 now: c_2 = 110 + 1.1×(100−150) = 55.

6.3 Adding prices and inflation

If prices are p_1, p_2, the constraint becomes $p_1c_1 + p_2c_2/(1+r) = m_1 + m_2/(1+r)$. Define inflation π by $p_2 = p_1(1+\pi)$. Setting $p_1 = 1$ (so $p_2 = 1+\pi$):

BUDGET CONSTRAINT WITH INFLATION π ($p_1 = 1$, $p_2 = 1+\pi$)

$$c_1 + \frac{1+\pi}{1+r} c_2 = m_1 + \frac{m_2}{1+r}$$

REARRANGED AS A LINE c_2 = (slope) × c_1 + (intercept)

$$c_2 = -\frac{1+r}{1+\pi} c_1 + \frac{m_1(1+r)+m_2}{1+\pi}$$

- With **no** inflation the slope is −(1+r).
- With inflation the slope is **−(1+r)/(1+π) = −(1+ρ)**, where **ρ (rho) is the real interest rate**.

Example: Real interest rate

Bank pays 10% (nominal), inflation is 5%. 1 + ρ = 1.10/1.05 = 1.0476, so real rate ρ ≈ **4.76%**. Your money grows 10%, but prices also rise 5%, so your *purchasing power* grows only about 4.76%.

Common mistake

Real rate is **not** exactly nominal − inflation (10% − 5% = 5%). That is a good approximation only when rates are small. The exact formula is (1+r)/(1+π) − 1.

PART 7

Asset markets & no-arbitrage

Why every price must equal a present value

7.1 Same safe return everywhere

If there is no uncertainty, **all assets must offer the same rate of return**. If one asset gave more, nobody would buy the other (so its price would fall until returns equalise).

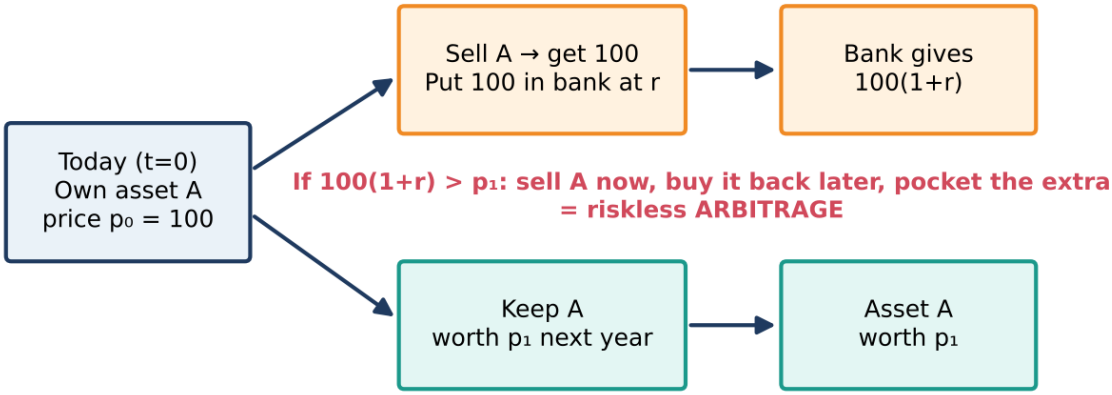
Take asset A (price p_0 today, known price p_1 next year, no dividends) vs a bank paying r. ₹1 in the bank → 1 + r. ₹1 in A → buys $1/p_0$ units, worth p_1/p_0 next year.

NO-ARBITRAGE CONDITION ⇒ PRICE = PRESENT VALUE

$$1 + r = \frac{p_1}{p_0} \quad \Longleftrightarrow \quad p_0 = \frac{p_1}{1+r}$$

7.2 What is arbitrage?

Riskless arbitrage = a way to make sure profit with zero risk and zero investment (buy cheap here, sell dear there). In a well-functioning market these chances vanish almost instantly because traders rush to exploit them, which moves prices.



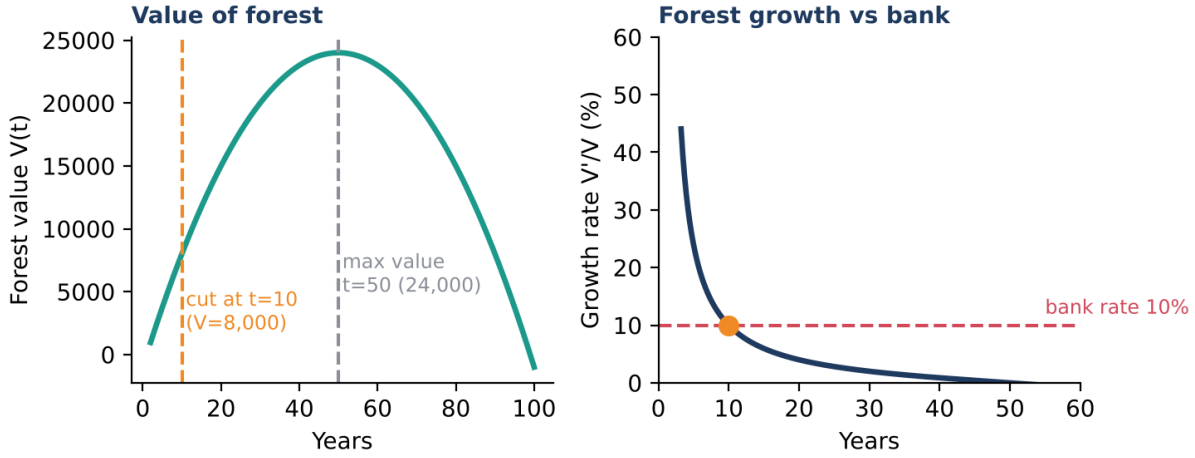
Example: Spotting arbitrage

Asset A costs ₹100 now and will certainly be worth ₹110 next year. Bank rate is 5%.
A returns 10% vs bank 5%: A is a bargain. Fair price should be 110/1.05 = **₹104.76**. Trade: borrow ₹100 at 5%, buy A, next year sell A for 110, repay 105 → **₹5 profit, no risk**. Everyone does this → demand for A rises → price rises to 104.76.

When the rule needs adjusting: assets that are harder to sell are less **liquid**; investors demand a higher return from them. The simple rule assumes the assets are otherwise identical.

7.3 Application: When should you cut down a forest?

A forest's value V(t) grows with time but the *growth rate* slows down. Meanwhile money in the bank earns a fixed rate. Rule: **cut when the forest's growth rate falls to the interest rate.** Before that, the trees earn more than the bank (keep growing); after that, the bank is better (cut and deposit).



Example: Your slide's numbers

V(t) = −1000 + 1000t − 10t². Value is highest at t=50 (V = 24,000), but that is **not** the best time to cut!

Growth rate = V′/V = (1000 − 20t)/(−1000 + 1000t − 10t²). Set it equal to 10%: solving gives t = 10 (the other root t=110 is meaningless). V(10) = 8,000.

Cut at t=10 and invest 8,000 at 10% for the remaining 40 years: 8,000 × 1.1^{40} ≈ **362,074** ≫ 24,000.

Lesson

Don't just maximise the asset's value. Compare its **rate of growth** with the best alternative return. Always hold the asset with the highest return at each moment.

7.4 Compounding more often, and continuous compounding

If interest is paid n times a year at rate r/n each time, ₹1 grows in T years to:

FROM n TIMES A YEAR → CONTINUOUS COMPOUNDING

$$(1+\tfrac{r}{n})^{nT} \longrightarrow e^{rT} \quad (n \to \infty)$$

Paid…	Formula (1 year, r = 6%)	₹1,000 becomes
Yearly	1000(1.06)	₹1,060.00
Monthly	$1000(1+0.06/12)^{12}$	₹1,061.68
Daily	$1000(1+0.06/365)^{365}$	₹1,061.83
Continuously	$1000 \cdot e^{0.06}$	₹1,061.84

More frequent payments help, but only slightly: the benefit levels off at e^{rT}. Present value with continuous discounting: PV = F(T)·e^{-rT}. Maximising it (set derivative to zero) gives **F′(T)/F(T) = r**: this is the forest rule proven formally.

PART 8

Perpetuities & annuities

Shortcuts for long streams of equal payments

Real investments pay many times: shares pay dividends for years, house loans have 200+ monthly instalments. Adding 200 discounted terms by hand is painful, so we have formulas for two common patterns (with a constant rate r):

Name	Pattern	Payments last…
Perpetuity	C, C, C, … starting at t=1	forever
Growing perpetuity	C_1, $C_1(1+g)$, $C_1(1+g)^2$, …	forever
Annuity	C, C, …, C (T payments)	T periods only

8.1 Perpetuity

Think of a magic bank account: deposit X today and it pays you ₹20 every year forever, keeping X intact. At 10%, the interest on X is 0.10 X = 20, so X = 200. That is exactly the formula:

PERPETUITY (first payment at t = 1)

$$PV_0 = \frac{C}{r}$$

Example: ₹20 a year forever at 10%

PV = 20/0.10 = **₹200**. (Proof from slides: 20/1.1 + 20/1.1² + … is a geometric series summing to 20/0.1.)

Watch the timing!

The formula assumes the **first payment is one period from now**. If the first payment is *today* (t = 0), then PV = C + C/r.

8.2 Growing perpetuity

Payments grow at g each period (e.g. dividends rising 5% a year). The first payment C_1 comes at t = 1.

GROWING PERPETUITY

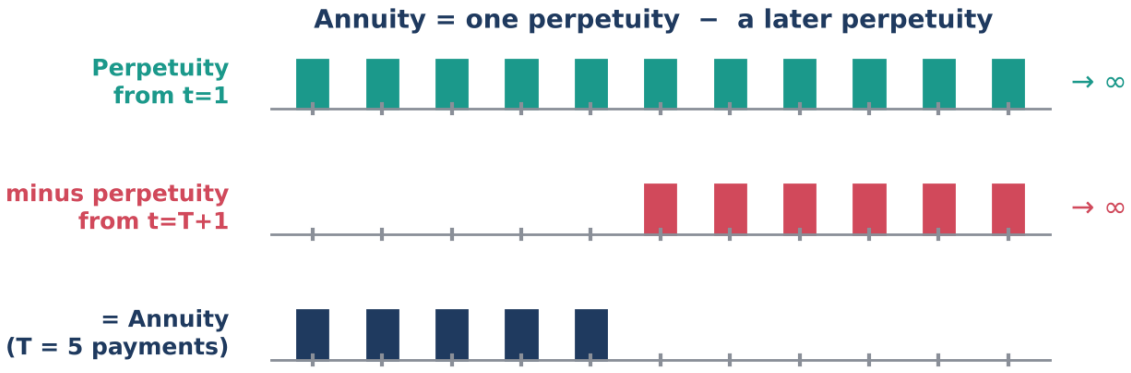
$$PV_0 = \frac{C_1}{r-g} \qquad (r > g)$$

Example: 20 next year, growing 5%, r = 10%

PV = 20/(0.10 − 0.05) = **₹400**. The growth makes it worth twice the flat perpetuity (200). We need r > g, or the value would be infinite.

8.3 Annuity

An annuity pays C for T periods and then stops. Trick: an annuity = a perpetuity starting now **minus** a perpetuity that starts after the annuity ends.



ANNUITY FORMULA (first payment at t = 1)

$$PV_0 = \frac{C}{r}\left[1 - \frac{1}{(1+r)^T}\right] = \frac{C}{r} - \frac{C}{r(1+r)^T}$$

Example: ₹50 for 3 years at 10%

Long way: 50/1.1 + 50/1.1² + 50/1.1³ = 45.45 + 41.32 + 37.57 = 124.34.
Formula: (50/0.10) × [1 − 1/1.1³] = 500 × 0.2487 = **₹124.34** ✔

8.4 Application: the quiz-show 'millionaire'

David wins a prize 'worth ₹1 million': 20 annual instalments of ₹50,000 starting one year from now, with interest at 10%.

$$PV = \frac{50{,}000}{0.10}\left[1 - \frac{1}{(1.1)^{20}}\right] \approx 425{,}678$$

The prize is worth only about **₹4.26 lakh today** (less than ₹5 lakh), so David's finance-student relative is right. The newspaper simply added up numbers from different dates, which is exactly the mistake time value of money warns us about.

PART 9

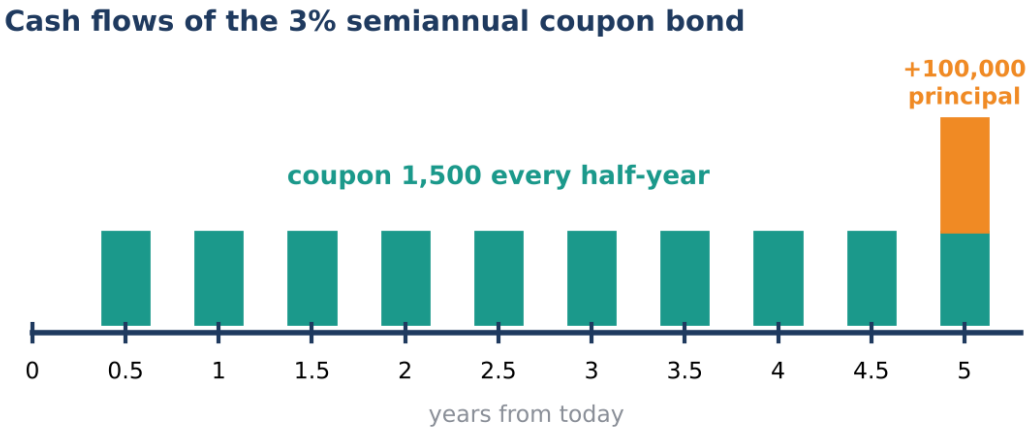
Application: pricing bonds

Putting annuity + discounting together

A **bond** is a claim sold by a firm or government. Two main kinds:

- **Zero-coupon bond**: pays one lump sum at maturity, nothing before. Price = PV of that one amount.
- **Coupon bond**: pays regular **coupons** and returns the **principal** (face value) at the end, along with the last coupon.

A **level-coupon bond** has the same coupon every time. It is named by (yearly coupons ÷ principal). ₹3,000 a year on ₹100,000 principal = a **'3% coupon bond'**. **The 3% is just a name, not the interest rate!**



9.1 Your slide's bond

Pays **1,500 every half-year for 5 years** plus **100,000 at year 5**. Market interest rate = 5% per year. Value today?

Step	What to do	Result
1	List payments: 1,500 at years 0.5, 1, ... 4.5; and 101,500 at year 5	10 cash flows
2	Discount each by $1/(1.05)^t$ (t in years)	e.g. $1{,}500/1.05^{0.5}$ for the first
3	Add all present values	**91,501.42**

Faster with the annuity formula: the half-year rate r_h solves $(1+r_h)^2 = 1.05 \rightarrow r_h = 2.47\%$.

Piece	Calculation	Value
10 coupons (annuity)	$1{,}500 \times [1 - 1/(1.0247)^{10}] / 0.0247$	13,148.81
Principal (lump sum)	$100{,}000 / (1.05)^5$	78,352.62
Bond price		**91,501.42**

Premium vs discount

Bond price (91,501) < principal (100,000) → the bond trades **at a discount**.
Bond price > principal → trades **at a premium**.
Why? This bond's coupon rate (3%) is below the market rate (5%), so buyers pay less than face value to make up the difference. If the coupon rate were above the market rate, it would trade at a premium. If equal → price = principal.

PART 10

Formula cheat-sheet

Revise from this page

Concept	Formula	Remember
Rate of return	$r = (C_1 + \text{Div} - C_0)/C_0$	= capital gain + dividend yield
Future value	$FV = C_0(1+r)^t$	compounding = interest on interest
Present value	$PV = C_t/(1+r)^t$	discounting = reverse of compounding
NPV	$NPV = C_0 + \Sigma C_t/(1+r)^t$	accept if > 0
Budget constraint	$c_1 + c_2/(1+r) = m_1 + m_2/(1+r)$	PV(spending) = PV(income)
Real rate	$1+\rho = (1+r)/(1+\pi)$	$\approx r - \pi$ when small
No-arbitrage	$1 + r = p_1/p_0 \Rightarrow p_0 = p_1/(1+r)$	price = present value
Optimal cutting time	$F'(T)/F(T) = r$	growth rate = interest rate
Continuous compounding	$FV = C \cdot e^{rT}$, $PV = C \cdot e^{-rT}$	limit of $(1+r/n)^{nT}$
Perpetuity	$PV = C/r$	first cash at t=1
Growing perpetuity	$PV = C_1/(r - g)$	need r > g
Annuity	$PV = (C/r)[1 - 1/(1+r)^T]$	T payments from t=1
Zero-coupon bond	$P = \text{Face}/(1+r)^T$	one lump sum
Coupon bond	$P = \text{annuity(coupons)} + \text{Face}/(1+r)^T$	premium if P > Face
Basis point	1 bp = 0.01%; 100 bp = 1%	

Top 6 exam traps

① Rates must match the period (half-year rate for half-year payments).
② Two-year return of 10%/yr is 21%, not 20%.
③ Perpetuity/annuity formulas value cash **one period before the first payment**.
④ In NPV include the upfront cost as a **negative** number at t=0.
⑤ Coupon rate ≠ interest rate.
⑥ Need r > g for the growing perpetuity.

PART 11

Practice problems

Solve first, then check the solutions that follow

Answers are on the next pages. Be honest with yourself: calculator out, no peeking!

Returns

1. You buy a share for ₹120. It pays a ₹4 dividend and you sell it for ₹130. Find the total rate of return, the capital gain %, and the dividend yield.

2. A bank cuts its rate from 8% to 5%. By how many percentage points and how many basis points?

Compounding

3. You invest ₹1,000 at 8% per year. What is it worth after 5 years? After 10 years?

4. A 2-year investment return is 44%. What constant annual rate produces this?

5. How many years to double your money at 7% per year?

6. If the annual rate is 6%, what is the equivalent six-month rate?

Present value

7. What is ₹5,000 to be received in 3 years worth today at 6%?

8. A bond pays ₹1,000 in one year. Find its price at 5% and at 8%. What does this tell you?

NPV

9. A project costs ₹500 now and pays ₹200, ₹250, ₹300 at the end of years 1, 2, 3. Discount rate 10%. Accept?

10. Same project, but discount rate 25%. Accept? What happened and why?

Perpetuity

11. An investment pays ₹60 a year forever starting next year. Value at 8%?

12. A stock will pay ₹60 dividend next year, growing 3% a year forever. Value at 8%?

Annuity

13. What is the present value of ₹1,000 a year for 5 years (first payment next year) at 10%?

14. A lottery pays ₹100,000 a year for 10 years, starting next year. At 8%, what is the prize really worth?

Bonds

15. Price a zero-coupon bond paying ₹1,000 in 10 years at a 4% interest rate.

16. A 3-year bond pays a ₹50 coupon each year and ₹1,000 principal at the end. Interest rate 5%. Price? Premium, discount or par?

Arbitrage

17. Asset A costs ₹100 and is certain to be worth ₹110 next year. The bank pays 5%. Is there arbitrage? What is A's fair price?

Compounding

18. ₹1,000 at 6% for 5 years: compare annual, monthly and continuous compounding.

Budget line

19. Income $m_1 = 100$ today, $m_2 = 110$ next year, $r = 10\%$. Find the maximum c_1 and the maximum c_2. Find c_2 if $c_1 = 150$.

Inflation

20. Nominal rate 10%, inflation 5%. Find the exact real interest rate and the slope of the budget line.

PART 11b

Solutions

With working, so you can see where marks come from

1. Return = (130 + 4 − 120)/120 = 14/120 = **11.67%**. Capital gain = 10/120 = **8.33%**. Dividend yield = 4/120 = **3.33%**. (8.33 + 3.33 = 11.67 ✔)

2. Fall of **3 percentage points = 300 basis points**.

3. $FV_5 = 1000 \times 1.08^5$ = **₹1,469.33**. $FV_{10} = 1000 \times 1.08^{10}$ = **₹2,158.92**.

4. $(1+r)^2 = 1.44 \rightarrow 1+r = 1.2 \rightarrow$ **r = 20%**.

5. $1.07^x = 2 \rightarrow x = \ln 2/\ln 1.07$ = **10.24 years** (Rule of 72: 72/7 ≈ 10.3 ✔).

6. $(1+r)^2 = 1.06 \rightarrow r = \sqrt{1.06} - 1$ = **2.96%** (not 3%).

7. $PV = 5000/1.06^3$ = **₹4,198.10**.

8. At 5%: 1000/1.05 = **₹952.38**. At 8%: 1000/1.08 = **₹925.93**. Higher interest rate → lower bond price.

9. PV = 181.82 + 206.61 + 225.39 = 613.82. NPV = 613.82 − 500 = **+₹113.82 → accept**.

10. PV = 160.00 + 160.00 + 153.60 = 473.60. NPV = **−₹26.40 → reject**. A higher discount rate shrinks future cash more, so the same project flips from good to bad. The discount rate is the 'hurdle' a project must beat.

11. PV = 60/0.08 = **₹750**.

12. PV = 60/(0.08 − 0.03) = 60/0.05 = **₹1,200**.

13. $PV = (1000/0.10) \times [1 - 1/1.1^5] = 10{,}000 \times 0.37908$ = **₹3,790.79**.

14. $PV = (100{,}000/0.08) \times [1 - 1/1.08^{10}]$ = **₹671,008.14**, far below the headline ₹10,00,000.

15. $P = 1000/1.04^{10}$ = **₹675.56**.

16. Price = $50/1.05 + 50/1.05^2 + 1050/1.05^3 = 47.62 + 45.35 + 907.03$ = **₹1,000.00**. Coupon rate (5%) equals the market rate, so it trades **at par** (price = principal).

17. Bank gives 5% but A gives 10% for sure, so yes, A is under-priced. Fair price = 110/1.05 = **₹104.76**. Arbitrage: borrow ₹100 at 5%, buy A, sell for 110 next year, repay 105 → ₹5 risk-free profit. Buying pressure pushes A's price up to ₹104.76.

18. Annual: 1000×1.06^5 = **₹1,338.23**. Monthly: $1000 \times (1+0.06/12)^{60}$ = **₹1,348.85**. Continuous: $1000 \times e^{0.3}$ = **₹1,349.86**. More frequent compounding → slightly more.

19. PV of income = 100 + 110/1.1 = 200 → **max c_1 = 200**. Save all: **max c_2 = 110 + 1.1 × 100 = 220**. If c_1 = 150: $c_2 = 110 + 1.1 \times (100 - 150) = 110 - 55$ = **55** (you borrowed 50, repay 55).

20. $1 + \rho = 1.10/1.05 = 1.0476 \rightarrow$ **ρ ≈ 4.76%**. Slope of budget line = $-(1+r)/(1+\pi)$ = **−1.0476** (flatter than −1.10 because inflation eats part of the gain from saving).

How did you do?

18–20 right: you've got it. 12–17: re-read Parts 4, 5 and 8 for the ones you missed. Under 12: redo the 'Example' boxes by hand, then retry. Finance is a skill you build by **doing** the arithmetic.